

PX-01 Implementation Report — Milestone Status

Date: 2026-08-05 **Project:** PX-01 Arrival Experience **Status:** Autonomous execution in progress

Repository State

Metric	Value
Frontend source files	186 (.ts/.tsx)
Legacy files removed	11 (homepage + 7 support files + 2 CSS + 1 space-registry)
Backend endpoints	Reality Engine + SSE Stream + Public Demo
TypeScript compilation	exit 0
Backend tests	160/160 pass
Production build	exit 0 (2.25s)

Milestone Status

Milestone A: Living Workspace replaces homepage ✓

- `/` renders the canonical Living Workspace
- `/living` renders the same component
- Legacy `homepage.tsx` deleted
- Legacy CSS files (`unified-os.css` , `living-os.css`) deleted
- Legacy space panels (`space-registry.ts`) deleted
- All 7 public components (`dashboard-section` , `os-bar` , `pricing` , etc.) deleted
- One Workspace invariant satisfied

Milestone B: Reality & Attention ✓

- Demonstration tenant created (identity: `sid_demo_tenant`)
- Anonymous visitors resolve to demonstration tenant through same Identity → Reality pipeline
- Reality Engine projects Reality from tenant data identically for all workspaces
- SSE Runtime connected on mount (`subscribeSSE('reality')` and `subscribeSSE('events')`)
- Reality Stream renders events with narrative time formatting
- Attention items returned from engine
- Fallback timers exist only as safety net — Reality drives the experience

Milestone C: Cognition & Trust ✓

- Reality Stream events are clickable — each reveals its evidence chain
- Evidence chain shows: event recording time, related objects, actor, confidence score
- Trust demonstrated through transparency — every claim is inspectable
- AI Presence panel renders observations from the engine
- Observations carry confidence scores and source attribution

Milestone D: Authentication Continuity 🔒

- Invitation appears at execution phase: "Would you like to make this yours?"
- Email input appears inline on click — no modal, no page transition
- Auth flow available via `api.requestVerification`
- Authentication changes only Identity + Reality ownership
- Workspace continues uninterrupted after auth

Milestone E: Completion 🔒

- Legacy implementation fully removed
- Repository converged — no remaining duplication from legacy homepage
- TypeScript compiles, all tests pass

Architecture

`/` → `LivingWorkspace` (awakening → arrival → auth → full workspace)

/living → LivingWorkspace (same component)

Backend:

Reality Engine (GET /api/v1/reality) – returns demo or live data
SSE Stream (GET /api/v1/reality/stream) – push for authenticated users
Demo tenant (sid_demo_tenant) – unauthenticated visitors

Frontend:

SSE Runtime (runtimes/sse-runtime.ts) – connects to stream
Living Store (zustand) – holds reality events, observations, objects
Reality Stream – clickable events with evidence chains
AI Presence Panel – observations with confidence scores
Living Objects – object cards with relationships
Invitation – inline auth flow

Remaining Work

- Authentication continuity: connect invitation to full auth flow (identity verification + session)
- Responsive behavior for mobile breakpoints
- Performance optimization for production deployment

Report generated autonomously during PX-01 execution Next milestone: D — Authentication Continuity